

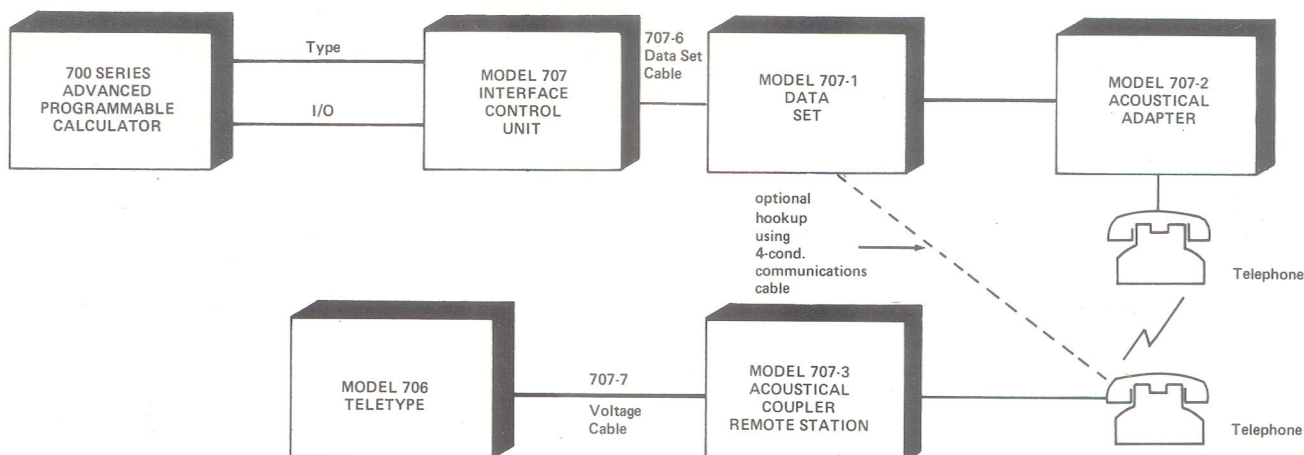
SERIES 706/707

This series of equipment enables any Series 700 Calculator user to utilize a teletype as an input/output writer for the calculator. Optional equipment in this series provides an unlimited number of equipment setup configurations. No limit is placed on the number of teletype terminals that can be interfaced to the calculator. Physical locations of the teletypes can be dictated by the user's needs

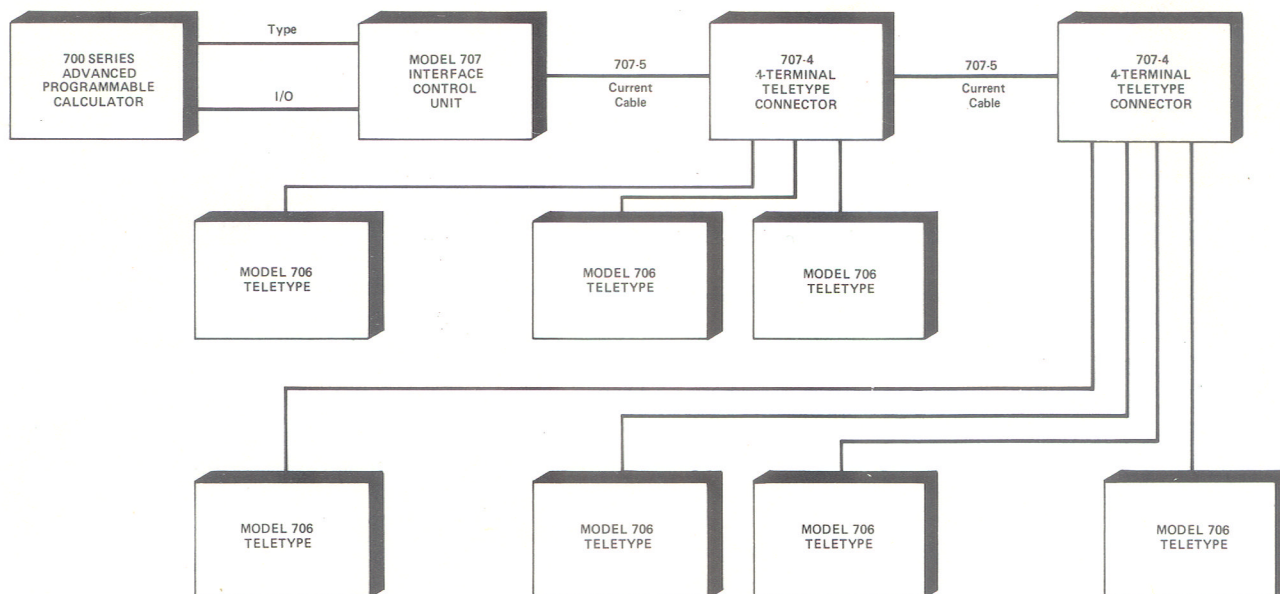
through the use of conventional telephone lines, special communication lines, or extension cables. By using a data set and acoustic couplers, communications between a teletype and calculator is simply a matter of dialing the data set number and cradling the telephone in an acoustic coupler. A few of the more common configurations are shown below.



BASIC TELETYPE CONFIGURATION



TYPICAL REMOTE TELETYPE CONFIGURATION



MULTIPLE TELETYPE CONFIGURATION

WANG SERIES 706/707 (cont.)

Wang Series 706/707 is comprised of nine items. These items are described briefly in the following paragraphs. Any user desiring the advantages provided by this equipment may either purchase the complete series or individual components, depending on his specific needs.

MODEL 707 INTERFACE CONTROL UNIT

The Model 707 converts the input/output codes of the 700 Series calculator to or from, respectively, the ASCII codes of the Model 706 Teletype.

Any number of teletypes can be interfaced to the calculator through the Model 707 by using T-connectors (see Model 707-4). If remote teletype operations are desired, the Model 707 can be connected to a data set (see Model 707-1 explanation) which uses telephone lines as the communications link.

Connections are made to both the calculator and teletype(s) through plug-in cables. Two cables link the interface to the rear panel of the calculator; another cable is used between the interface and teletype or T-connector. The connection to the I/O calculator connector is used for Group I and Group II, and input function. The connection to the typewriter connector on the calculator is used for calculator output and converts parallel Selectric Bail code to serial ASCII code for driving the teletype.

MODEL 707-4 FOUR-TERMINAL TELETYPE CONNECTOR

A Model 707-4 is used when connecting more than one Model 706 Teletype to a Model 707 Interface Control Unit. Current Cable 707-5 is used to couple the connector to the teletypes and interface unit.

Four or less teletypes can be connected to each Model 707-4. If more than four teletypes are required, a second Model 707-4 can be connected to one of the channels of the first level Model 707-4 using a current cable. With a 2-level arrangement, seven teletypes can be used. Connectors can be chained in this fashion to accommodate any number of teletypes. If desired, the current cable can be extended to a length of 1,000 feet.

Busy-line lockout is not provided in the Model 707-4. Any teletype that is connected and turned on will act as a repeater or interrupt communication.

MODEL 707-1 DATA SET

The Model 707-1 is available for users who desire calculator access from remote teletype locations. Either of two methods can be used to link the remote station to the calculator configuration. One method uses telephone lines for transmission of data; if this arrangement is used, an acoustic adaptor and coupler are required (see Model 707-2&3 explanations). This configuration causes the remote station to act as the control center with the calculator center becoming the slave station. Conventional telephone service is necessary for this configuration. Transmissions are made by dialing the telephone number of the remote station and placing the telephone headset into the Model 707-2 Acoustic Adaptor.

The second method uses a direct access arrangement whereby 4-conductor cabling is wired from the telephone at the remote station to the Model 707-1 Data Set at the calculator location.

A Model 707-6 Data Set Cable, supplied with the Model 707-1, is used to connect the data set to the Model 707 Interface Control Unit. An integral cable on the acoustic adaptor connects to the other end of the data set. A Western Electric Data Set 103A together with the 804B Data Auxiliary Set may be substituted for the Model 707-1 and 707-2.

MODEL 707-6 DATA SET CABLE

This 6-foot cable is used to connect the Model 707-1 Data Set to the Model 707 Interface Control Unit (see Data Set explanation).

MODEL 707-2 ACOUSTIC ADAPTER

The Model 707-2 is required whenever conventional telephone lines are used as the link between a remote teletype site and the calculator site. It is connected directly to the Model 707-1 Data Set and functions as the coupling device between the telephone headset and the data set. A Western Electric Data Auxiliary Set, 804B may be substituted for this adapter.

MODEL 707-3 ACOUSTIC COUPLER (REMOTE STATION)

The Model 707-3 is used at remote teletype locations to interface the teletype to the telephone. Connection to the remote telephone is made by placing the telephone's headset on the acoustic coupler. The Model 707-7 Voltage Cable, supplied with the Model 707-3, connects the acoustic coupler to the teletype.

MODEL 707-7 VOLTAGE CABLE

Refer to the Model 707-3 explanation.

WANG SERIES 706/707 (cont.)

MODEL 707-5 CURRENT CABLE

This cable is supplied with the Model 706 and is used to connect the Teletype to the Model 707 Interface Control Unit. It is also used as the interconnecting cable between Model 707-4 Teletype Connectors and the Model 707. It is furnished in a 12-foot length to fit most equipment setup applications. However, a user requiring a longer cable may increase this length to 1,000 feet using 4-conductor cable.

MODEL 706 TELETYPE

The Model 706 is an unmodified ASR 33TTY Teletype. All connections to the teletype are made at terminal 151411 on the rear panel. A Model 707-5 Current Cable, furnished with each Model 706 connects the teletype to the Model 707 Interface Control Unit.

MODES OF OPERATION:

700 to Teletype

Write Mode — function same as 701 Output Writer.

Example:

The X-register of the 700 has the following number:

—123.45600000

The following 700 commands are given:

WRITE 03 05

The teletype will print

—123.45600

Write Alpha Mode — function same as 701 with additional commands of teletype punch on and off with set tab and clear tab codes.

Example:

Assume that the following commands are given on the 700:

WRITE ALPHA, 01 08, END ALPHA

The teletype will perform a carriage return and line feed.

Teletype to 700 (Run Mode)

Group-I 15X. The teletype will respond to the 15X command. Control will pass to the teletype. 700 will then be in the Run Mode. The 700 will then only respond to the teletype characters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, ., ↑(set exp.), —(CHS), ←(clear display), Control A(GO), Control B(S&R00), Control C(S&R01), Control D(S&R02). Rubouts will be ignored. Any other character will be considered illegal and will automatically generate a "resume" to the 700 if the illegal character was preceded by a legal character.

Example:

Assume that the following 700 commands are given:

GROUP 1, 15 05

These two commands pass control to the teletype leaving the 700 in the Run Mode. If the characters 1, 2, & 3 are typed on the teletype, the X-register of the 700 will display

+123

If a Control C is now typed on the teletype, the 700 will begin executing subroutine 01.

Teletype to 700 (Learn Mode)

Group II 15X. The teletype will respond to the 15X command. Control will pass to the teletype. The 700 will then be in the Learn Mode. All of the alpha and numeric characters may be learned into the 700 as well as Control U (Write A), and Control V (End A), Control H (upper case), Control L (lower case).

Example:

Assume that the following 700 commands are given:

GROUP 2, 15 06

These two commands pass control to the teletype leaving the 700 in the Learn Mode. We can now learn into the 700 from the teletype any alpha-numeric character (e.g., Control U(Write Alpha), W, A, N, G, Control V(End Alpha), Control A(resume to 700). The word "WANG" will now be in the memory of the 700.

The following 700 codes apply to both Group I and Group II operations.

A 700 XX code of 15 00 will generate an X ON (Reader on) and control will pass to the tape reader.

A 700 XX code of 15 01 will generate a 1 to the teletype. Control will pass to the teletype.

A 700 XX code of 15 02 will generate a 2 to the teletype. Control will pass to the teletype.

A 700 XX code of 15 03 will generate a 3 to the teletype. Control will pass to the teletype.

A 700 XX code of 15 04 will generate a 4 to the teletype. Control will pass to the teletype.

700 XX codes of 15 05 through 15 15 will just pass control to the teletype.

Teletype to 700 I/O Bus (700 placed in Learn Mode by Group 2 command or 700 to teletype when used as an output writer).
700 output codes used for teletype operations are for the most part identical to 701 Selectric codes.

U.C.	CHARACTER	L.C.	TELETYPE ASCII CODE Upper Case	TELETYPE ASCII CODE Lower Case	SERIES 700 CODE	U.C.	CHARACTER	L.C.	TELETYPE ASCII CODE Upper Case	TELETYPE ASCII CODE Lower Case	SERIES 700 CODE
	A		100 0001	Same	01 12	&		6	010 0110	011 0110	03 04
	B		100 0010		02 00	*		7	010 0110	011 0111	03 13
	C		100 0011		02 12	(		8	010 1010	011 1000	03 12
	D		100 0100		02 13	)		9	010 1000	011 1001	03 00
	E		100 0101		02 05			0	010 1001	011 0000	03 01
	F		100 0110		00 14	+		=	010 1011	010 1101	00 00
	G		100 0111		00 15	:			010 1011	011 1101	00 06
	H		100 1000		02 01	;		;	011 1010	011 1011	00 13
	I		100 1001		01 04	"			010 0010	010 0111	01 05
	J		100 1010		00 07	?		/	010 1100	Same	00 12
	K		100 1011		02 04			!	011 1111	010 1111	01 06
	L		100 1100		03 15				010 0000	010 0001	00 09
	M		100 1101		01 15				000 1010	Same	01 07
	N		100 1110		02 06		SPACE		000 1000		00 02
	O		100 1111		01 09		INDEX (LF)				01 10
	P		101 0000		00 05		SHIFT UP (Control H)		000 1000		01 03
	Q		101 0001		00 04		SHIFT DOWN (Control L)				
	R		101 0010		01 13		CR/LF		000 1100		01 02
	S		101 0011		01 01						
	T		101 0100		02 07		PUNCH ON (Control R)		000 1101		01 08
	U		101 0101		02 14		PUNCH OFF (Control T)		000 1010		00 10
	V		101 0110		01 14		WRITE ALPHA (Control U)		001 0100		00 11
	W		101 0111		01 00		END ALPHA (Control V)		001 0101		04 12
	X		101 1000		02 15		RESUME (Control A)		001 0110		04 13
	Y		101 1001		00 01				000 0001		05 14
	Z		101 1010		03 07						
	I		011 0001		02 09						
@		2	110 0000	011 0010	03 06						
#		3	010 0011	011 0011	03 14						
\$		4	*010 0100	011 0100	03 09						
%		5	010 0101	011 0101	03 05						

Teletype to 700 I/O Bus (700 placed in Run Mode by Group 1 Command)

CHARACTER (COMMAND)	TELETYPE ASCII CODE	SERIES 700 CODE	CHARACTER (COMMAND)	TELETYPE ASCII CODE	SERIES 700 CODE
0	011 0000	07 00	←(CL. DISP.)	010 1101	07 11
1	011 0001	07 01	↑(SET EXP.)	101 1111	07 15
2	011 0010	07 02	Control A	101 1110	07 10
3	011 0011	07 03	(GO)	000 0001	05 14
4	011 0100	07 04	Control B	000 0010	00 00
5	011 0101	07 05	(S&R 00)	000 0011	00 01
6	011 0110	07 06	Control C		
7	011 0111	07 07	(S&R 01)		
8	011 1000	07 08	Control D	000 0100	00 02
9	011 1001	07 09	(S&R 02)		
.	010 1110	07 12			

The last four codes will generate an X OFF in the teletype in a numeric character. In this latter case an illegal code will generate a
addition to the 700 generated codes. All other codes except rub-out
are illegal. All illegal codes will be ignored except when preceded by
GO to the 700 and an X OFF to the teletype. Rub-outs will just
be ignored.